

2 Echidnas are mammals that live in Australia and New Guinea.

Fig. 2.1 shows an echidna.



Fig. 2.1

Scientists analysed the milk produced by female echidnas and identified a protein that they named EchAMP. The scientists predicted that EchAMP may have antibacterial properties.

The scientists tested the effect of EchAMP on the bacterium *Escherichia coli*.

1. 100 *E. coli* cells were added to each well on a cell culture plate with 96 wells.
2. A treatment solution that contained EchAMP was added to each well on the plate.
3. A chemical that causes living *E. coli* cells to fluoresce was added to each well.
4. The plate was incubated at 37°C.
5. Every hour for 7 hours, the fluorescence emitted by the *E. coli* on the plate was recorded as a measure of *E. coli* population growth.
6. Steps 1–5 were repeated eight times.

The scientists also carried out two control experiments.

- A negative control experiment repeated the procedure (steps 1–6), but the treatment solution did not contain EchAMP.
- A positive control experiment repeated the procedure (steps 1–6), but the treatment solution contained an antibiotic called bacitracin instead of EchAMP.

(a) Identify the **independent** variable in this investigation.

..... [1]



- (b) The scientists standardised the temperature and the initial number of *E. coli* cells.

State **two other** variables that the scientists should standardise in this investigation.

1

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2

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[2]

- (c) (i) Explain why the scientists included the negative control experiment in their investigation.

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..... [1]

- (ii) Explain why the scientists included the positive control experiment in their investigation.

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..... [1]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

(d) Some of the results are shown in Table 2.1.

Table 2.1

time/hours	mean fluorescence/arbitrary units (au)		
	EchAMP	negative control	positive control
0	10	10	10
1	11	11	11
3	15	15	15
6	20	35	20
7	100	140	70

(i) Use Table 2.1 to complete the graph in Fig. 2.2 by:

- plotting the three results at 7 hours
- adding axis labels
- completing the key.

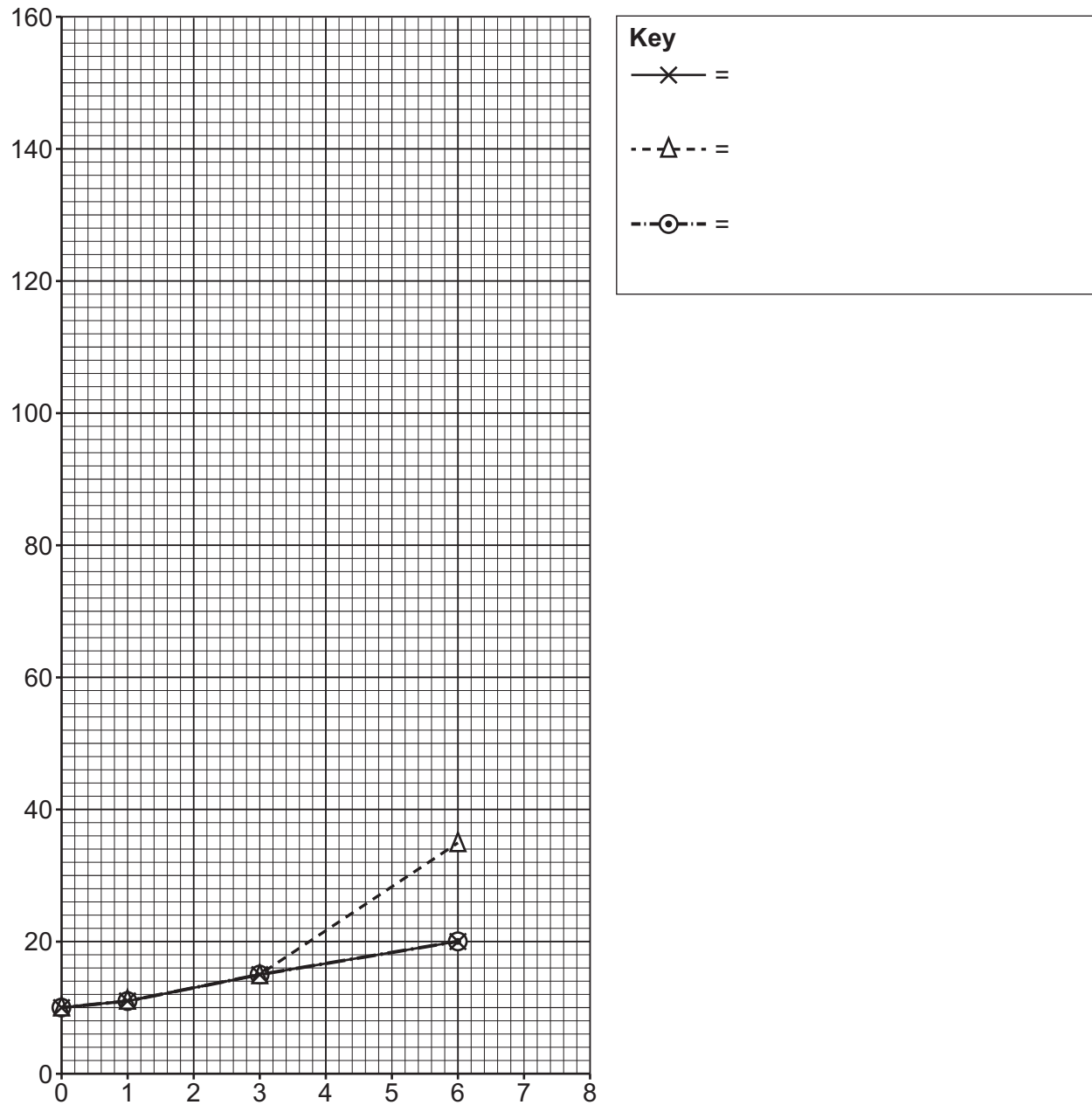


Fig. 2.2

[2]

- (ii) Use Table 2.1 and Fig. 2.2 to compare the effect of EchAMP and bacitracin on the population growth of *E. coli*.

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..... [2]

(e) The scientists used t -tests to analyse the results.

The scientists compared the fluorescence emitted by *E. coli* after seven hours when exposed to:

- the negative control
- EchAMP.

(i) State a null hypothesis for this t -test.

.....

 [1]

(ii) For *E. coli* exposed to the negative control, after seven hours:

- the mean of nine fluorescence measurements was 140 au
- the sample standard deviation was 8 au.

For *E. coli* exposed to EchAMP, after seven hours:

- the mean of nine fluorescence measurements was 100 au
- the sample standard deviation was 10 au.

The formula for calculating a t -test is:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

key to symbols

$\bar{x}$ = mean

s = sample standard deviation

n = sample size (number of observations)

Calculate a value of t for these data.

Show your working and state your answer to **four** significant figures.

$t =$ [3]



(iii) The t value calculated by the scientists was significant.

After reading the scientific paper published by the scientists, a student wrote the conclusion:

EchAMP would make a good treatment for bacterial infections in the human digestive system.

Suggest **four** reasons why this conclusion might **not** be valid.

- 1
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- 2
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- 3
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- 4
-
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[4]

[Total:17]